STRUCTURE,FUNCTIONS, PROPERTIES & OXYGEN- DISSOCIATION CURVE OF HEMOGLOBIN

LEARNING OBJECTIVES

At the end of lecture the student should be able to know;

- The synthesis of hemoglobin
- The Structural details of hemoglobin molecule specially its quaternary structure
- The T and R forms of Hemoglobin
- The oxygenation of hemoglobin molecule
- The Hemoglobin – oxygen dissociation curve and factors affecting it
- The Bohr's effect
- Compare the hemoglobin and myoglobin molecule with respect to structure and function

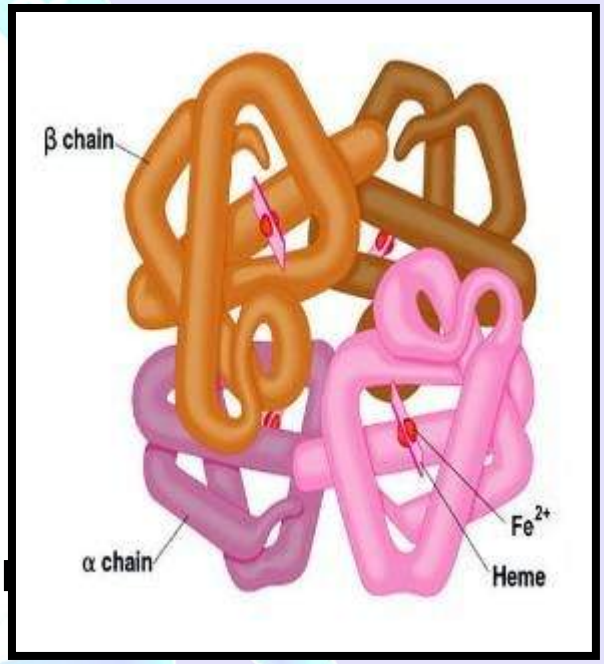
HEMOGLOBIN

- Hemoglobin is an oligomeric metalloprotein / chromoprotein

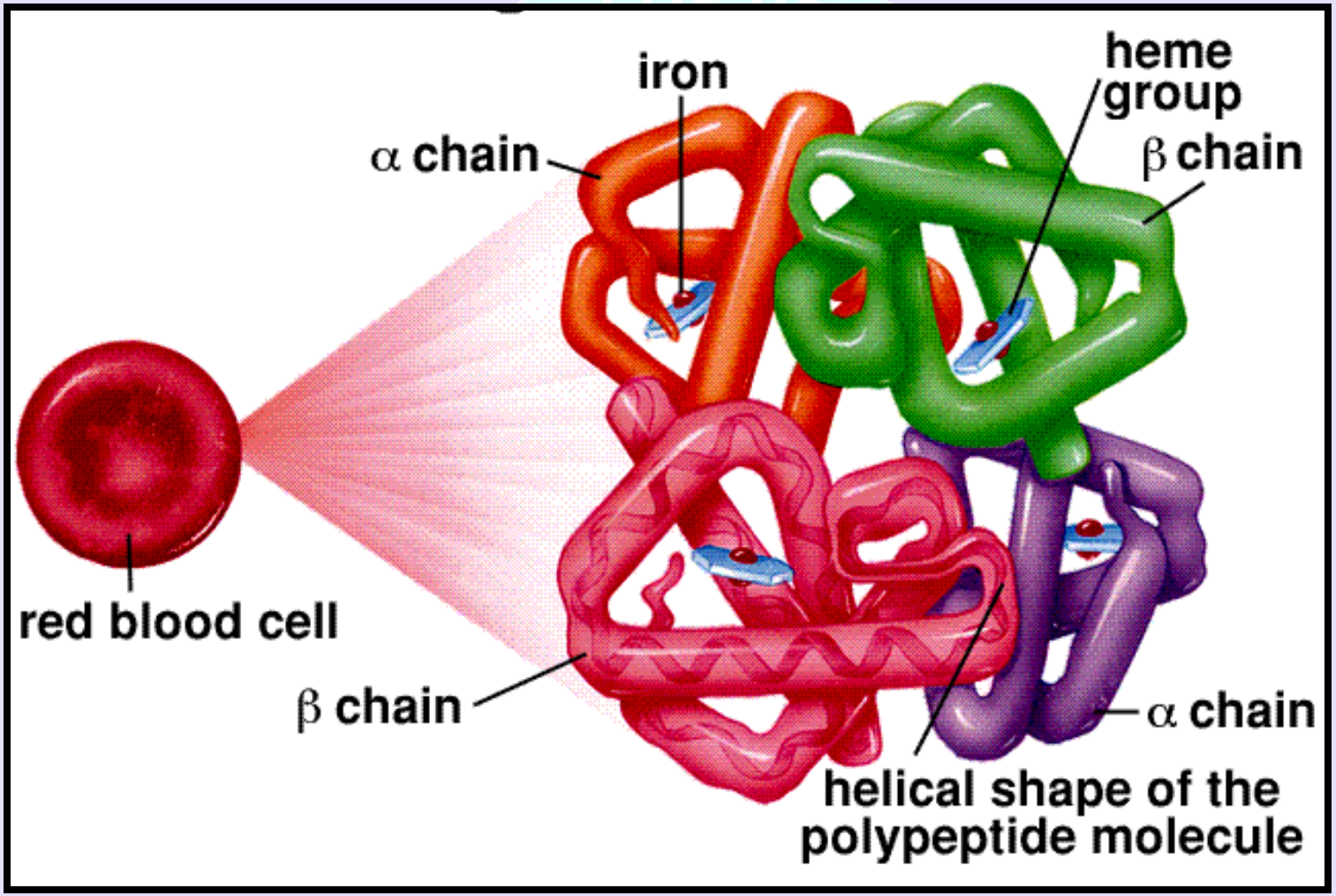
- It consists of four polypeptide chain, each with its own Heme

  i.e.

  Hemoglobin = 4 Heme + 4 Globin chains

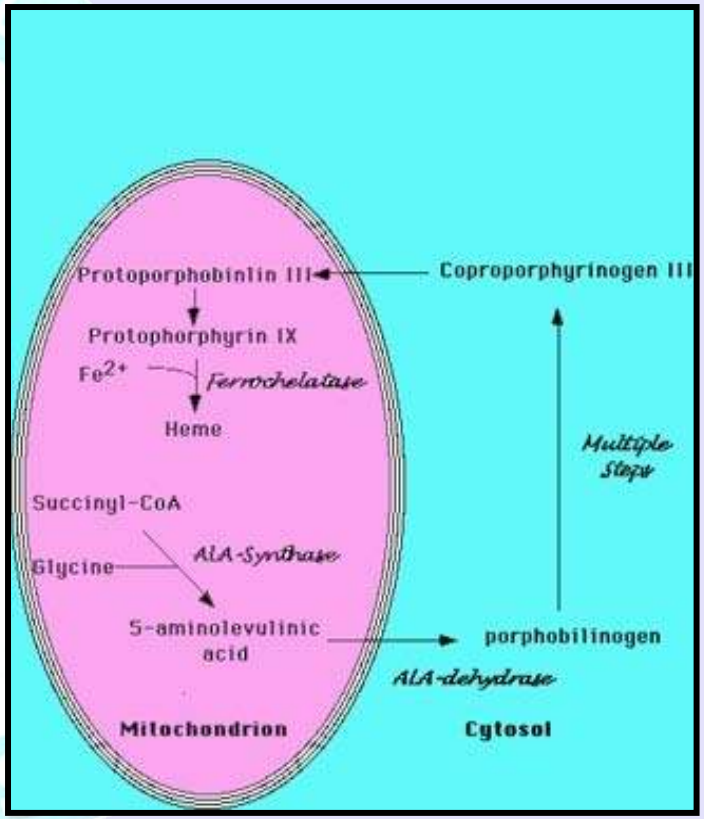


HEMOGLOBIN = **4** HEME + **4** GLOBIN CHAINS



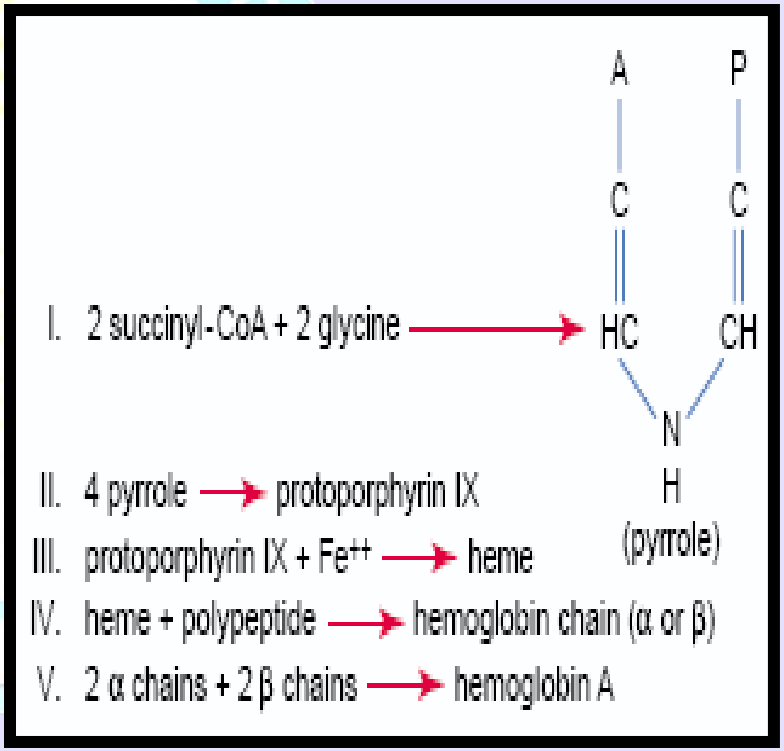
SYNTHESIS OF HEMOGLOBIN

- Synthesis of hemoglobin begins in the proerythroblasts and continues even into the reticulocyte stage of the red blood cells
- The process begins in the mitochondrion with the condensation of succinyl-CoA and glycine to form 5-aminolevulinic acid.
- succinyl-CoA, then , binds with glycine to form a pyrrole molecule.
- In turn, four pyrroles combine to form protoporphyrin IX, which then combines with iron to form the *heme* molecule.



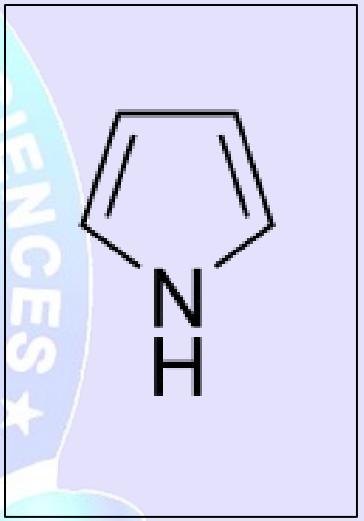
Synthesis of Hemoglobin…contd.

- Finally, each heme molecule combines with a long polypeptide chain, a *globin synthesized by* ribosomes, forming a subunit of hemoglobin called a *hemoglobin chain .*
- *Each chain has a* molecular weight of about 16,000; four of these in turn bind together loosely to form the whole hemoglobin molecule.



STRUCTURE OF HEME

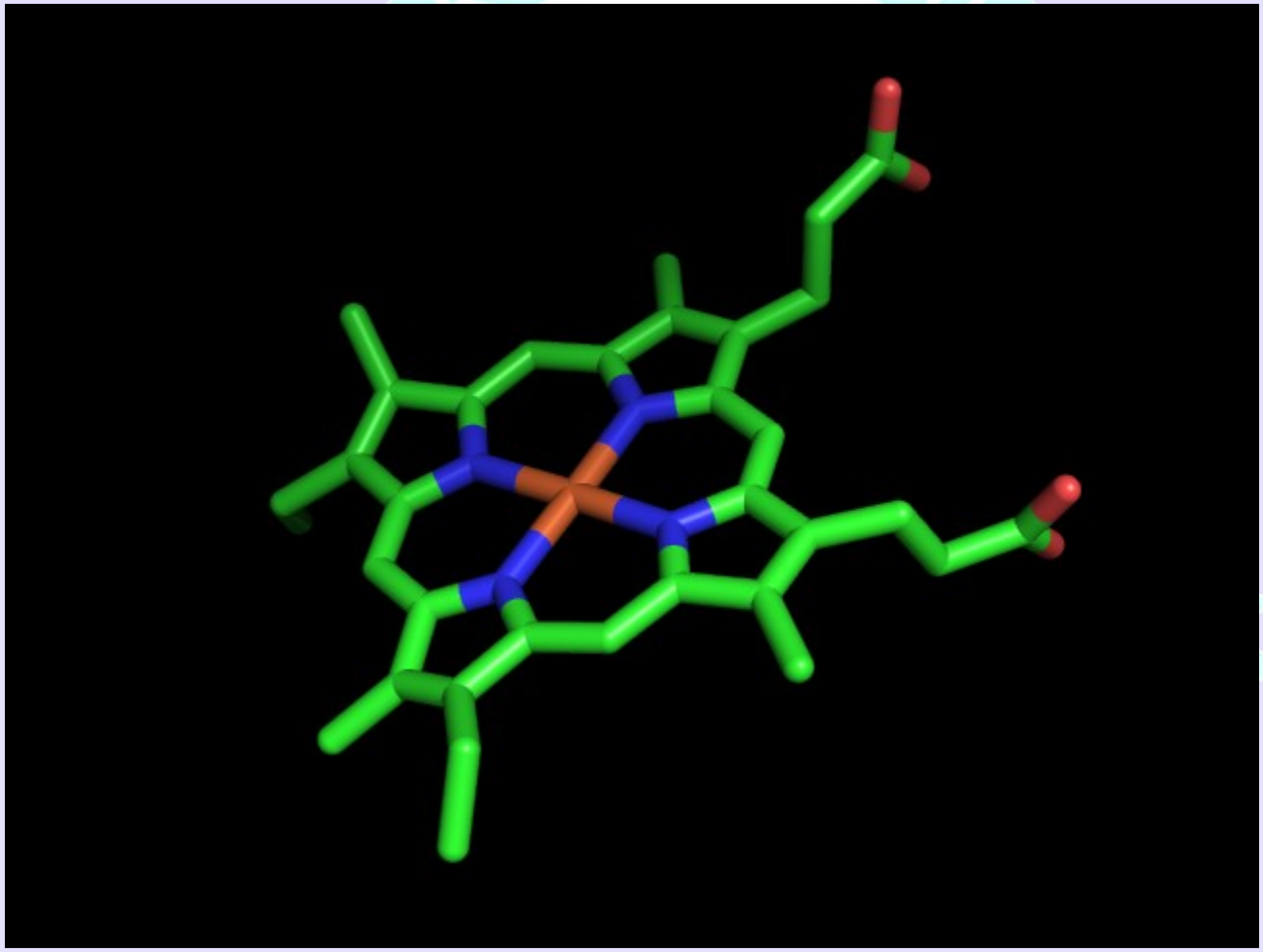
- Heme is a cyclic tetrapyrrole i.e. consists of four molecules of pyrrole.



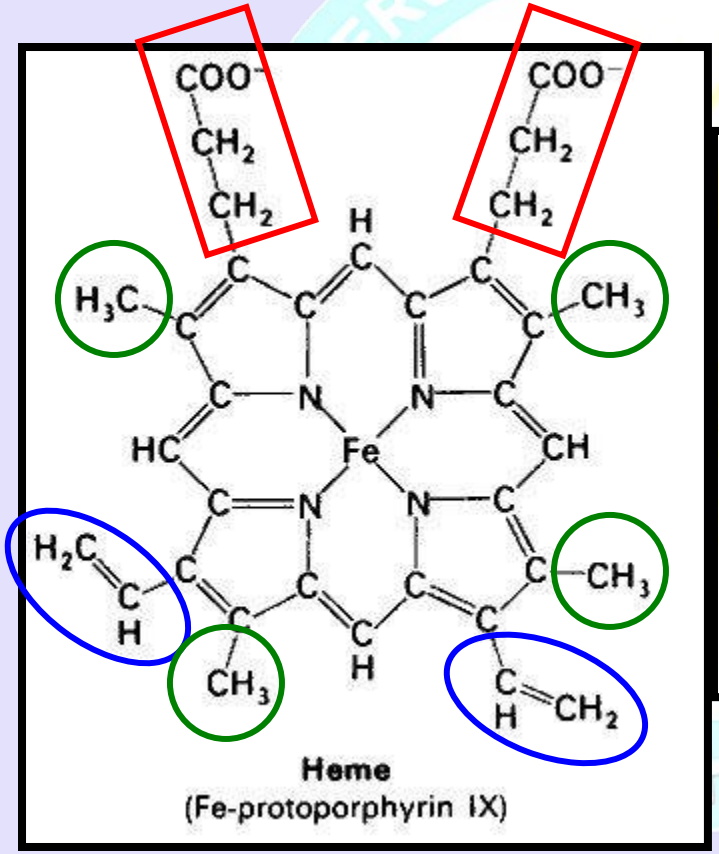
- This imparts a red color Pyrrole ring

- There are methyl (M), vinyl (V) and proprionate (Pr) groups attached to it

STRUCTURE OF HEME



STRUCTURE OF HEME



Propionate

Methyl

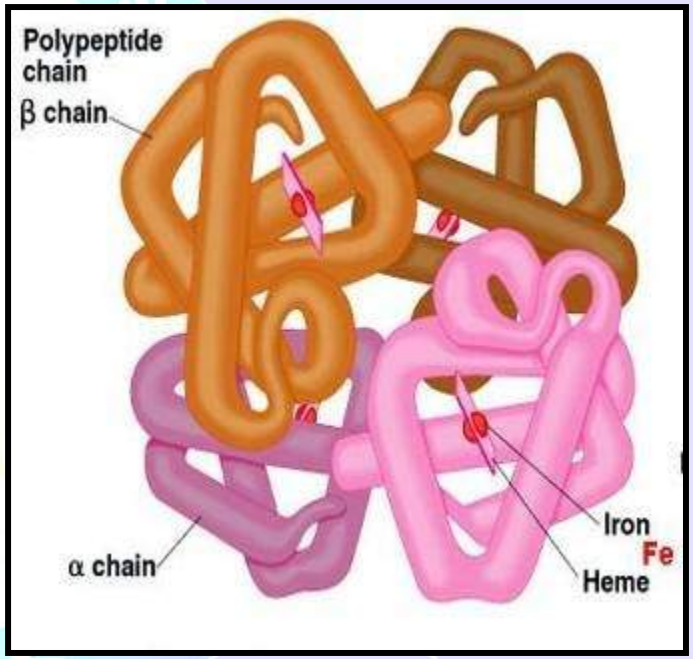
Vinyl

GLOBIN CHAINS

- There are four globin chains in each molecule of adult hemoglobin (**Hb-A**)
- These are designated as **α** and **β**
- There are two **α** (α_1 α_2) and two **β** (β_1 β_2) chains
- These four chains form two dimers
  i.e. $\boldsymbol{\alpha_1 \beta_1}$ and $\boldsymbol{\alpha_2 \beta_2}$

GLOBIN CHAINS

- **Hb-A** (adult hemoglobin) consists of
  - β_1, β_2 (two beta chains)
  - α_1, α_2 (two alpha chains)



NON-PATHOLOGICAL VARIANTS OF HEMOGLOBIN

In the embryo;

- Gower 1 ($\zeta_2\varepsilon_2$)
- Gower 2 ($\alpha_2\varepsilon_2$)
- Hemoglobin Portland ($\zeta_2\gamma_2$)

In the fetus:

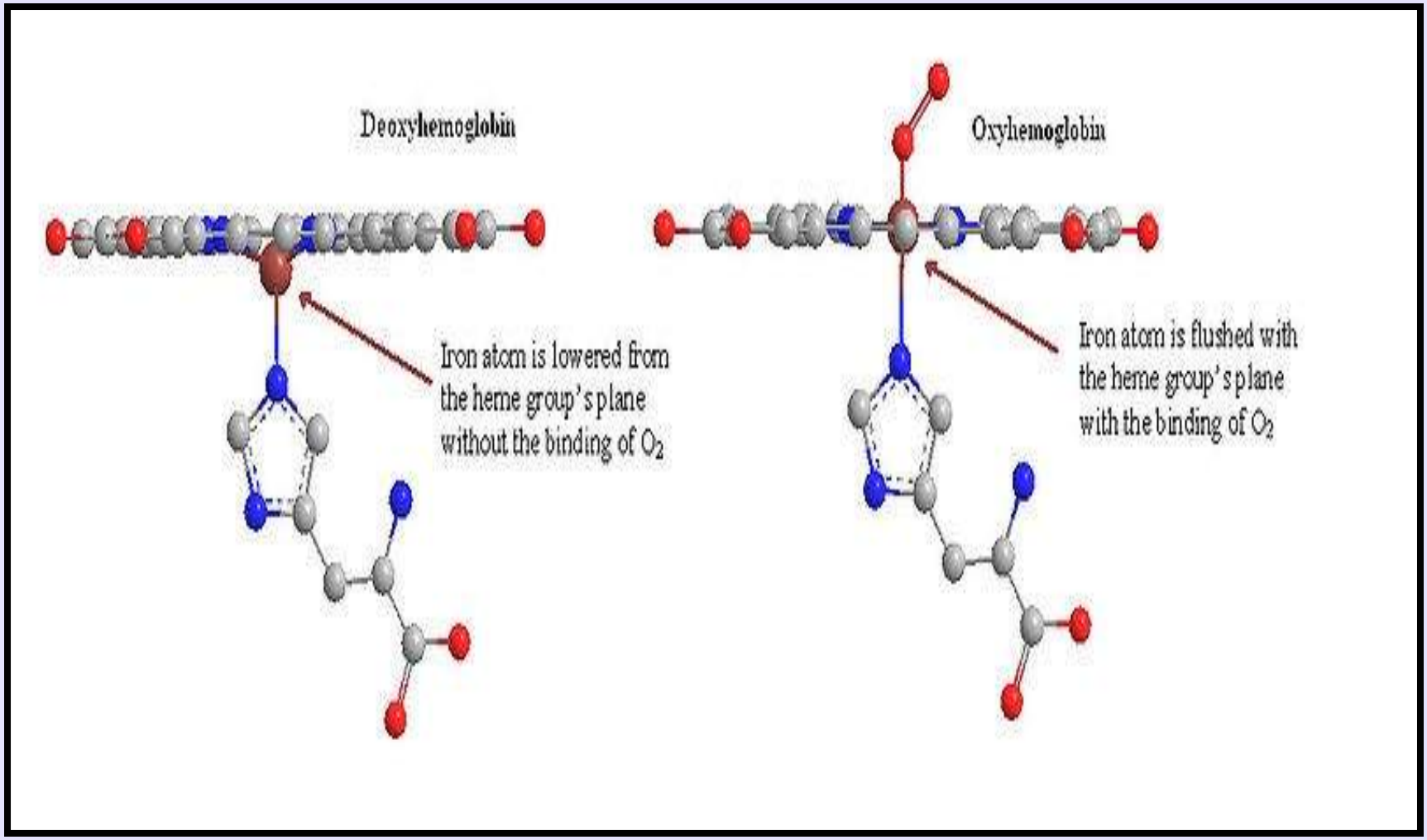
- Hemoglobin F ($\alpha_2\gamma_2$)

In adults:

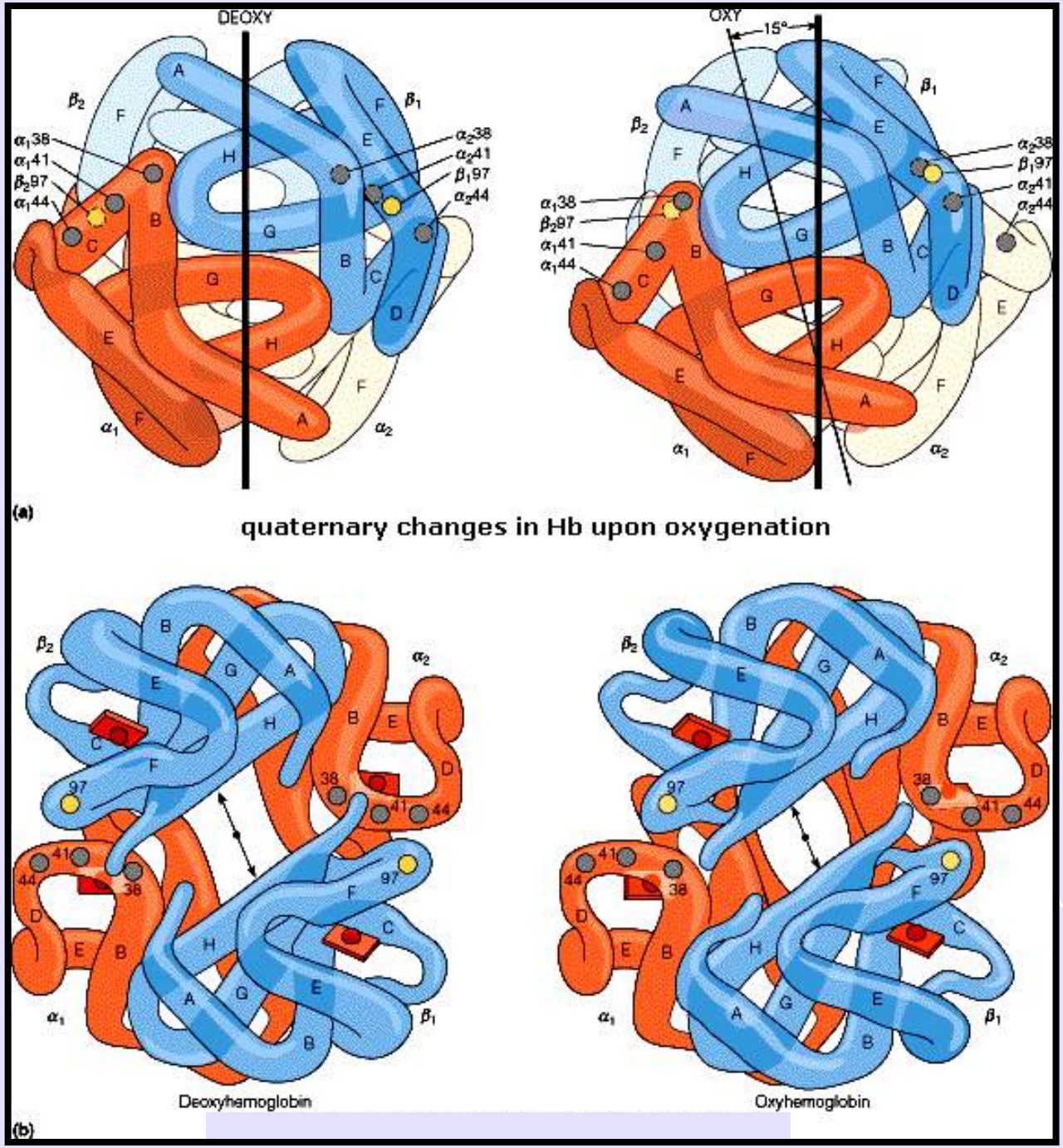
- Hemoglobin A($\alpha_2\beta_2$) The most common with a normal amount over 95%
- Hemoglobin A_2($\alpha_2\delta_2$) - δ chain synthesis begins late in the third trimester and in adults, it has a normal range of 1.5-3.5%
- Hemoglobin F($\alpha_2\gamma_2$) - In adults Hemoglobin F is restricted to a limited population of red cells called F-cells. However, the level of Hb F can be elevated in persons with sickle-cell disease and beta-thalassemia

QUATERNARY STRUCTURE OF HB

- Hemoglobin tetramer is composed of two identical dimers, $\alpha_1\beta_1$ and $\alpha_2\beta_2$
- The two globin chains within each dimer are held tightly together by inter chain hydrophobic interactions between **α** and **β** subunits.
- The two dimers are held together primarily by polar bonds and able to move with respect to each other.
- The weaker interactions between these mobile dimers result in two different relative positions in **Deoxyhemoglobin** compared to **Oxyhemoglobin**


Deoxyhemoglobin
Iron atom is lowered from the heme group's plane without the binding of O_2
Oxyhemoglobin
Iron atom is flushed with the heme group's plane with the binding of O_2

T OR TAUT FORM	R OR RELAXED FORM
• This is the De-oxy form of hemoglobin • The two **αβ** dimers are tightly interact trough ionic and hydrogen bonds and constrains the movement of dimers • This is a **low oxygen affinity** form of hemoglobin	• This is the Oxy form of hemoglobin • Binding of oxygen to Hb causes rupture of ionic & hydrogen bonds b/w dimers and have more freedom of movement • This is a **high oxygen affinity** form of hemoglobin


DEOXY
OXY
15°
(a)
quaternary changes in Hb upon oxygenation
(b)
Deoxyhemoglobin
Oxyhemoglobin

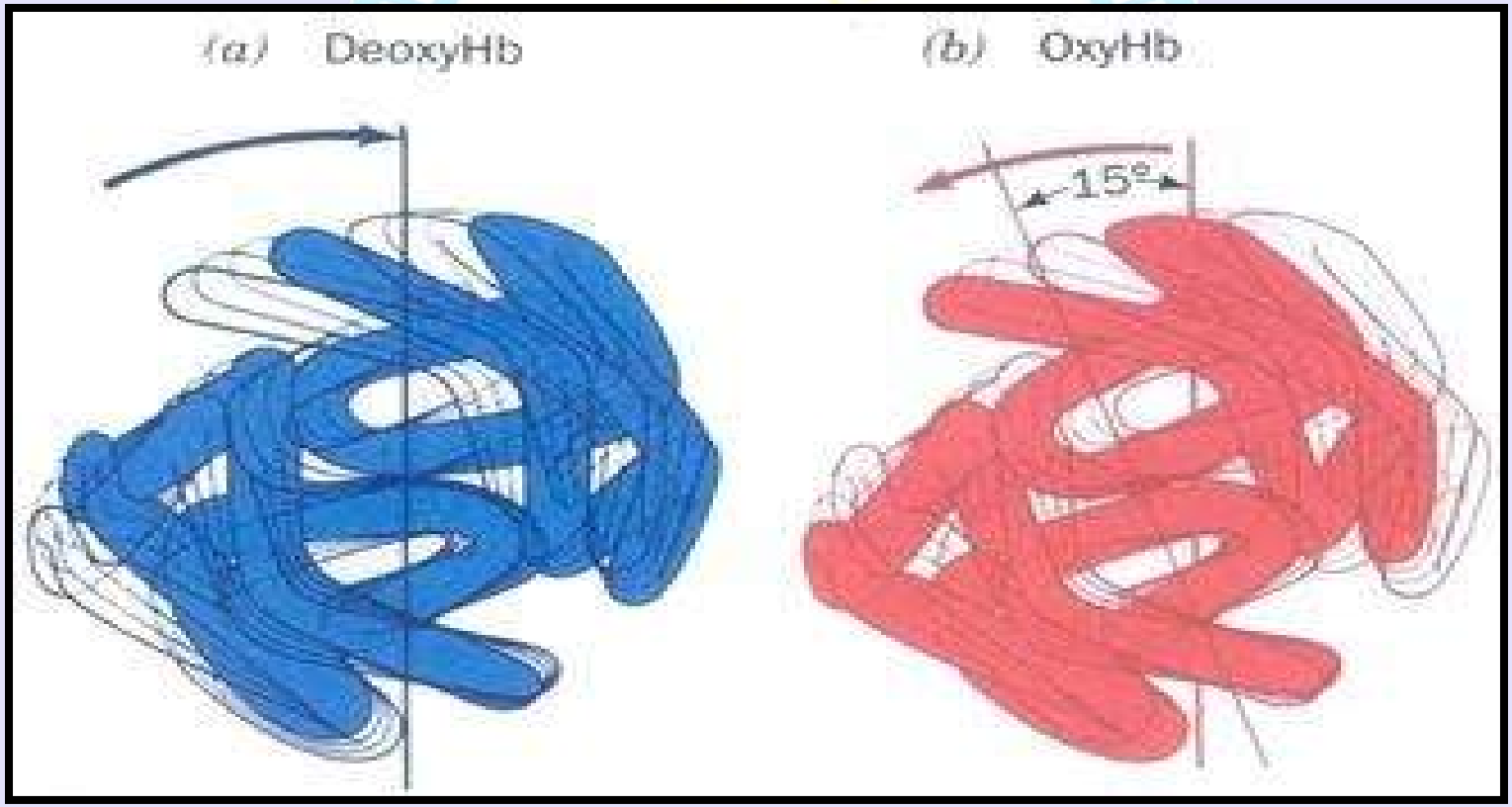
T AND **R** FORMS OF HEMOGLOBIN

T or **taut** or **tense** form

- The **deoxygenated** form

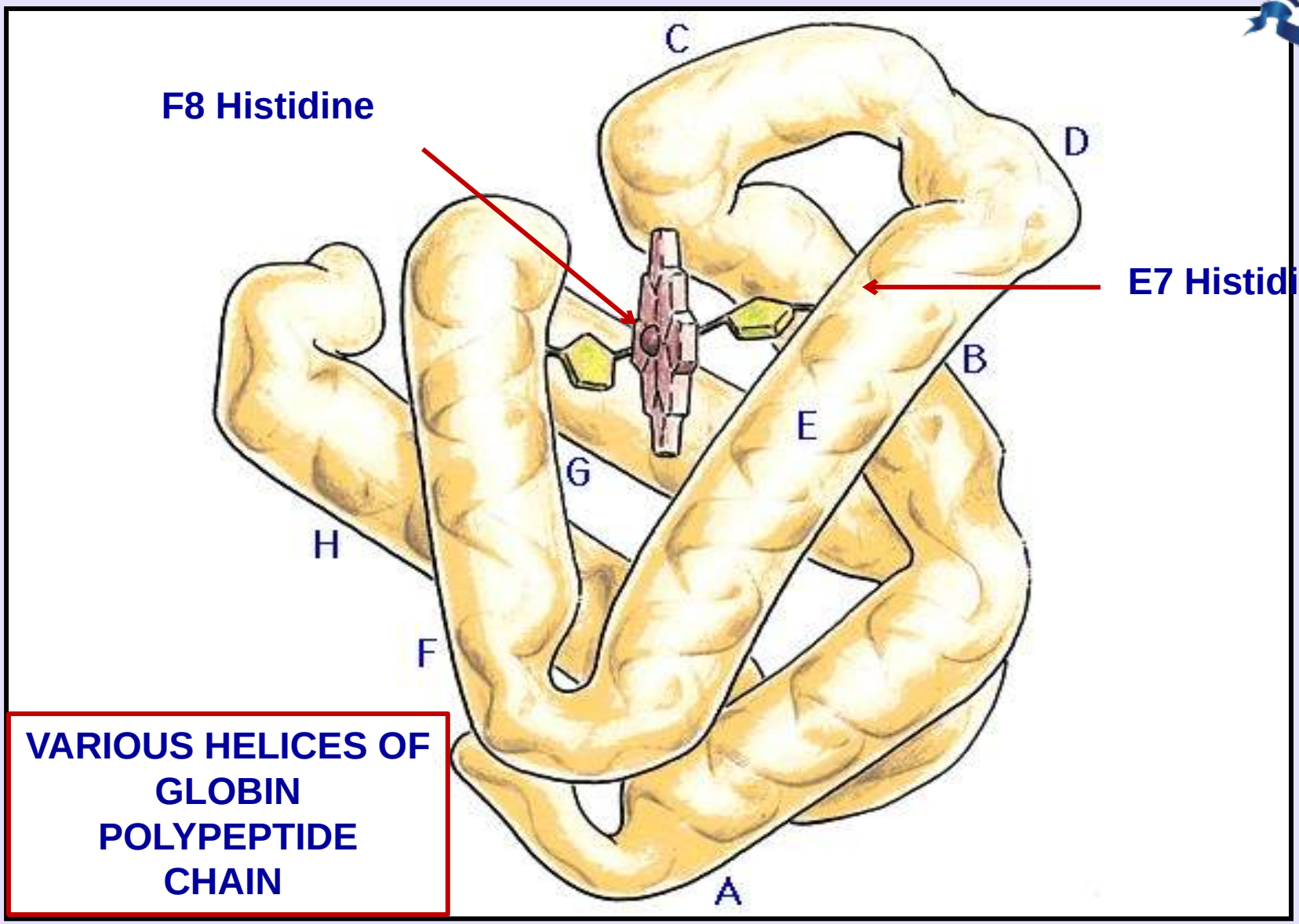
R or **relaxed** form

- The **oxygenated** form

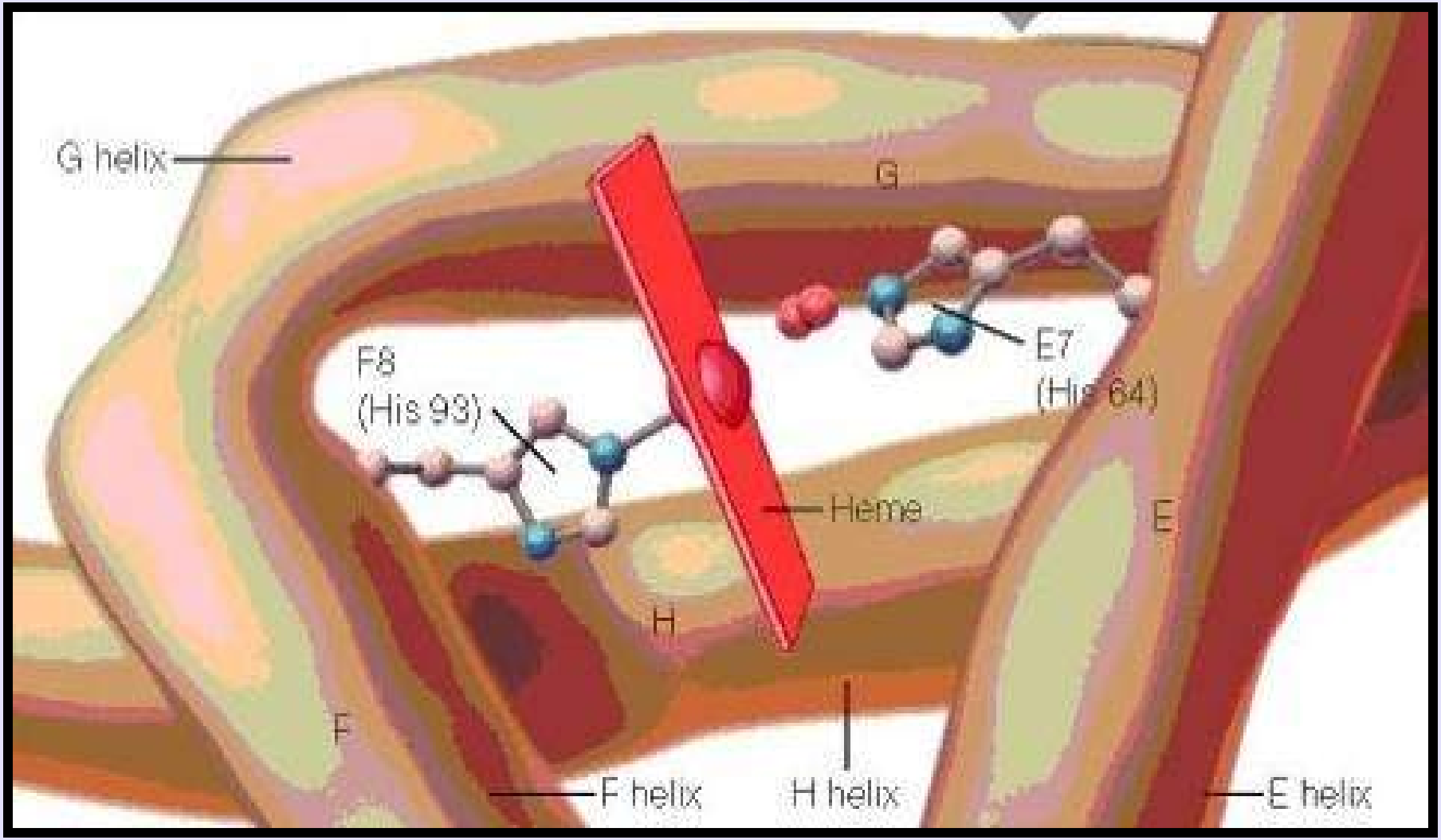


PROXIMAL (F8) & DISTAL (E7) HISTIDINE

- Hemoglobin has a relatively hydrophilic surface and hydrophobic interior.
- Polar amino acids are located almost exclusively on the exterior surface of globin polypeptide chain while the hydrophobic amino acids are buried within the interior.
- The only exception to this are two histidine residues termed as **proximal histidine (F8)** and **distal histidine (E7)**
- They play indispensible role in heme pocket and function in oxygen binding

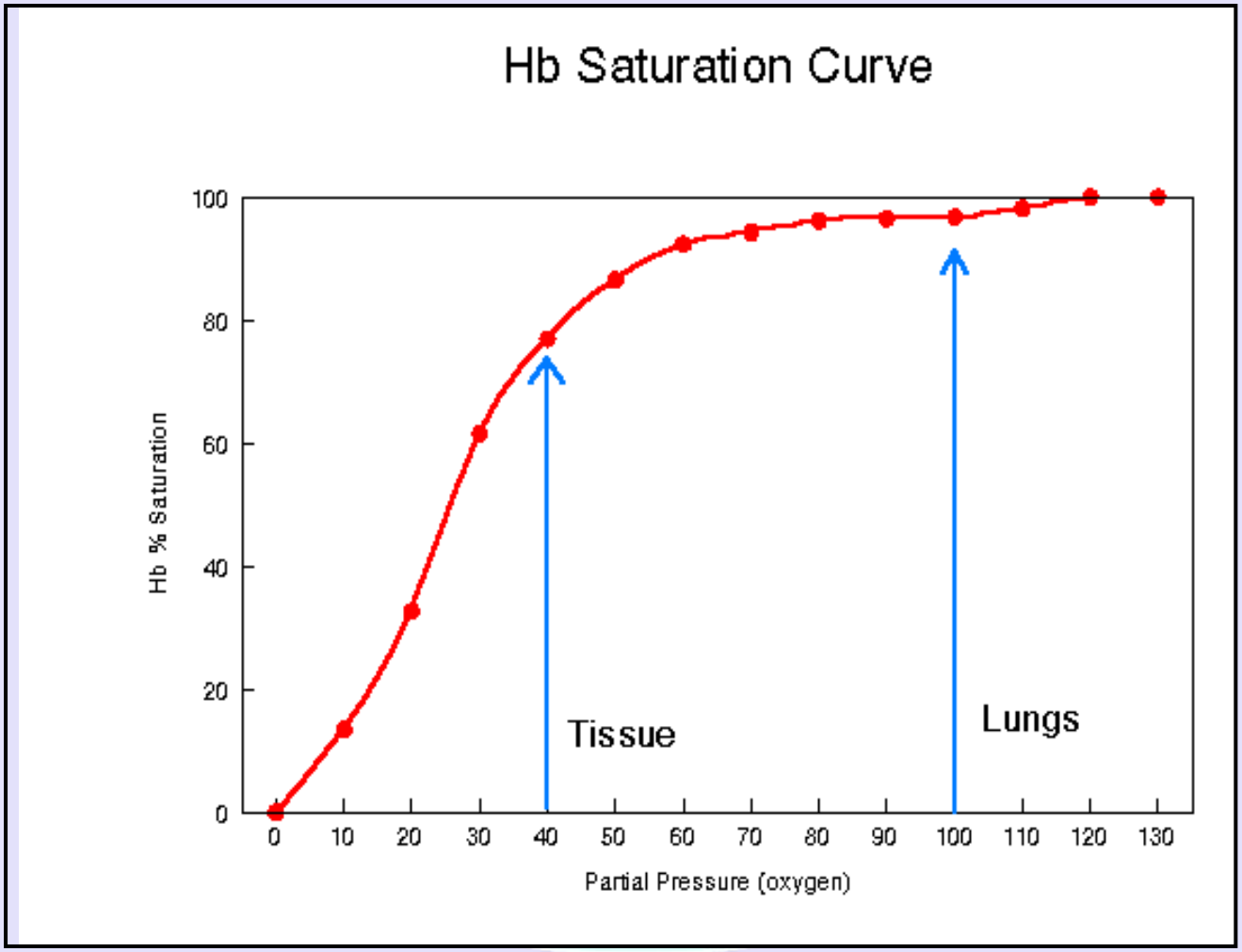


VARIOUS HELICES OF GLOBIN POLYPEPTIDE CHAIN


G helix
G
F8
(His 93)
E7
(His 64)
Heme
E
H
F
F helix
H helix
E helix

OXYGENATION OF HEMOGLOBIN

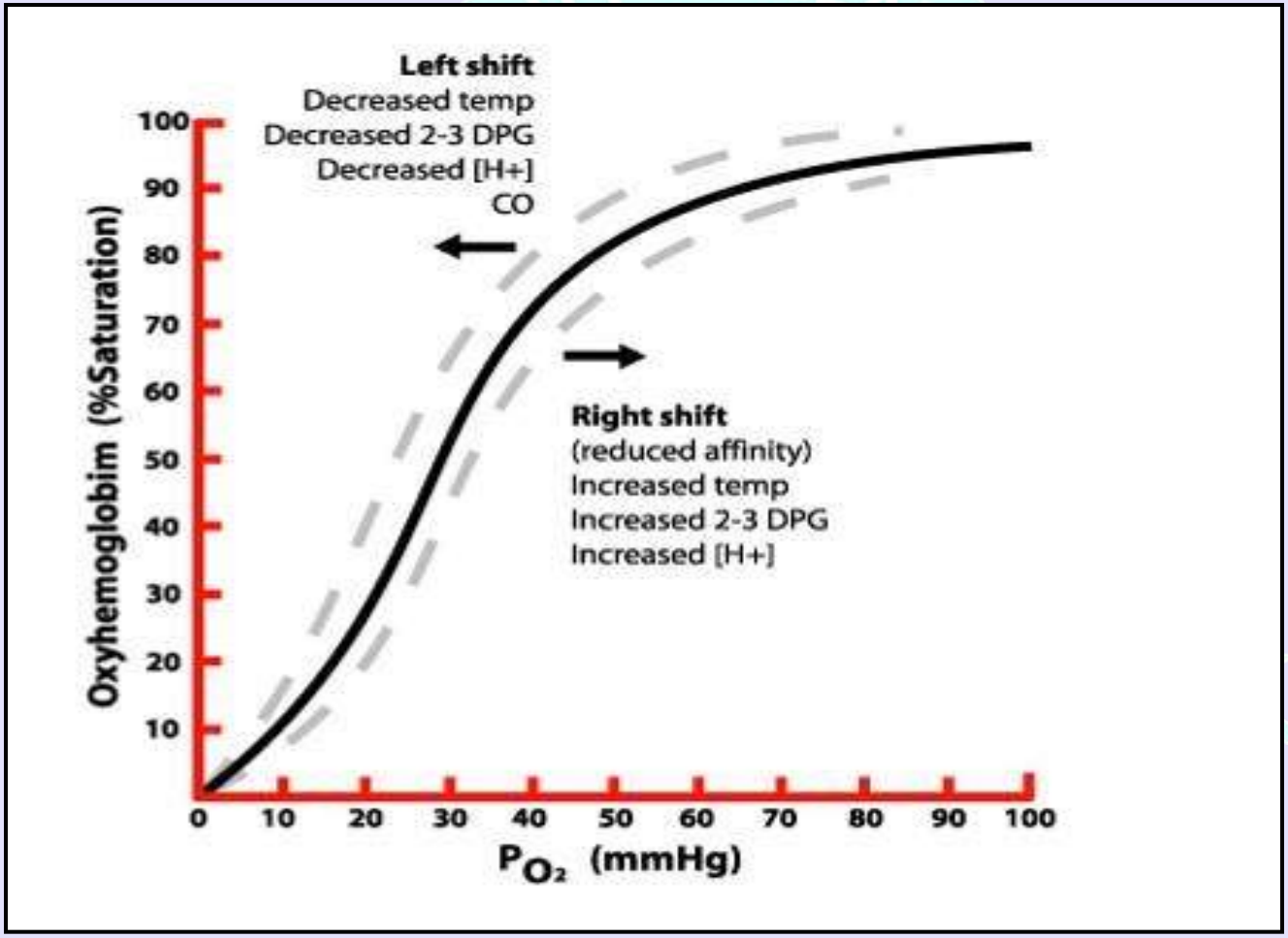
- Hemoglobin molecule can bind four **O_2** molecules (one per heme)
- Hb exhibits **cooperative binding kinetics** i.e. if **O_2** is already present, binding of subsequent **O_2** molecules occurs more easily
- This permits to bind maximum quantity of **O_2** at lungs (PO_2 = 100 mm Hg) and to deliver a maximum **O_2** at peripheral tissues (PO_2 = 20 mm Hg)
- This is shown by a sigmoid curve of oxygen dissociation of hemoglobin


Hb Saturation Curve
Hb % Saturation
100
80
60
40
20
0
0
10
20
30
40
50
60
70
80
90
100
110
120
130
Partial Pressure (oxygen)
Tissue
Lungs

OXYGENATION OF HEMOGLOBIN

- The oxygen binding characteristics of hemoglobin change in response to binding of various allosteric modulator viz:
  - The partial pressure of O2
  - pH of the surrounding medium
  - Presence of 2,3-diphosphoglycerate

FACTORS AFFECTING HB-SATURATION CURVE



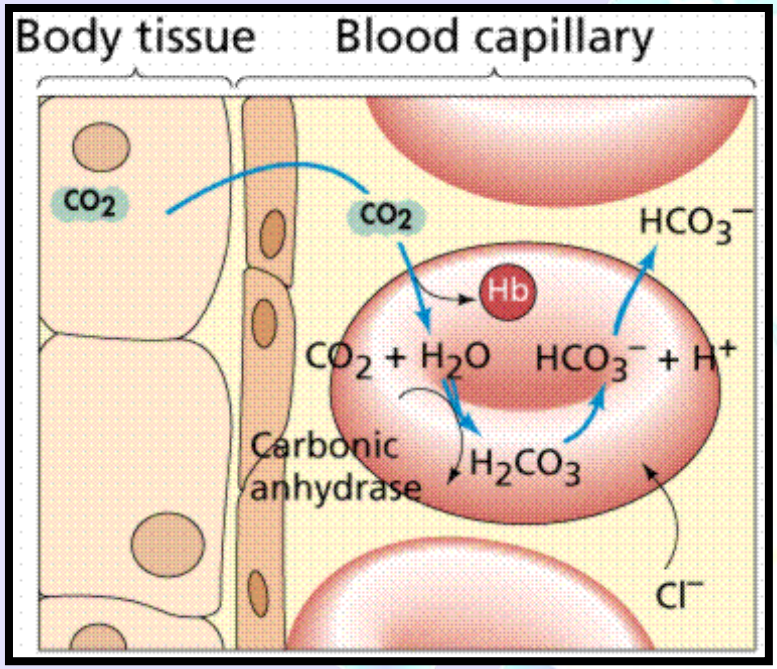
FUNCTIONS OF HEMOGLOBIN

- Transport of **O_2** from lungs to tissues
- Transport of **CO_2** and **H^+** from tissues to the lungs for excretion
- Hemoglobin is also found outside red blood cells and their progenitor lines. Other cells that contain hemoglobin include the A9 dopaminergic neurons in the substantia nigra, macrophages, alveolar cells, and mesangial cells in the kidney. In these tissues, hemoglobin has a non-oxygen-carrying function as an antioxidant and a regulator of iron metabolism.

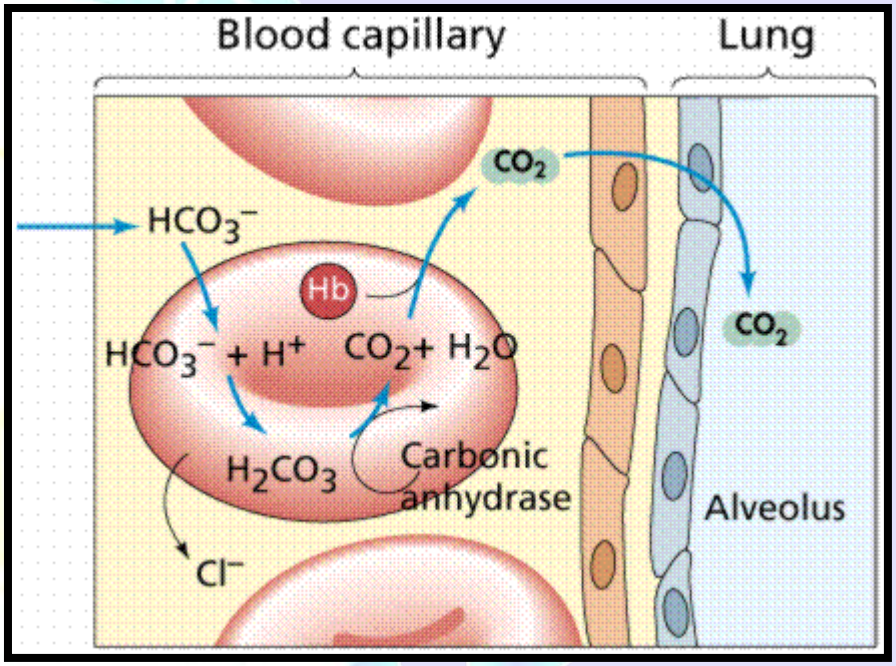
LOADING AND UNLOADING OF OXYGEN

- Deoxyhemoglobin binds one proton for every two molecules of oxygen released
- The slight lower pH in the tissues stabilizes the T state and enhances the O_2 delivery.
- In lungs the process reverses, as O_2 binds to deoxyhemoglobin protons are released and combine with bicarbonate to form carbonic acid.

AT TISSUE LEVEL

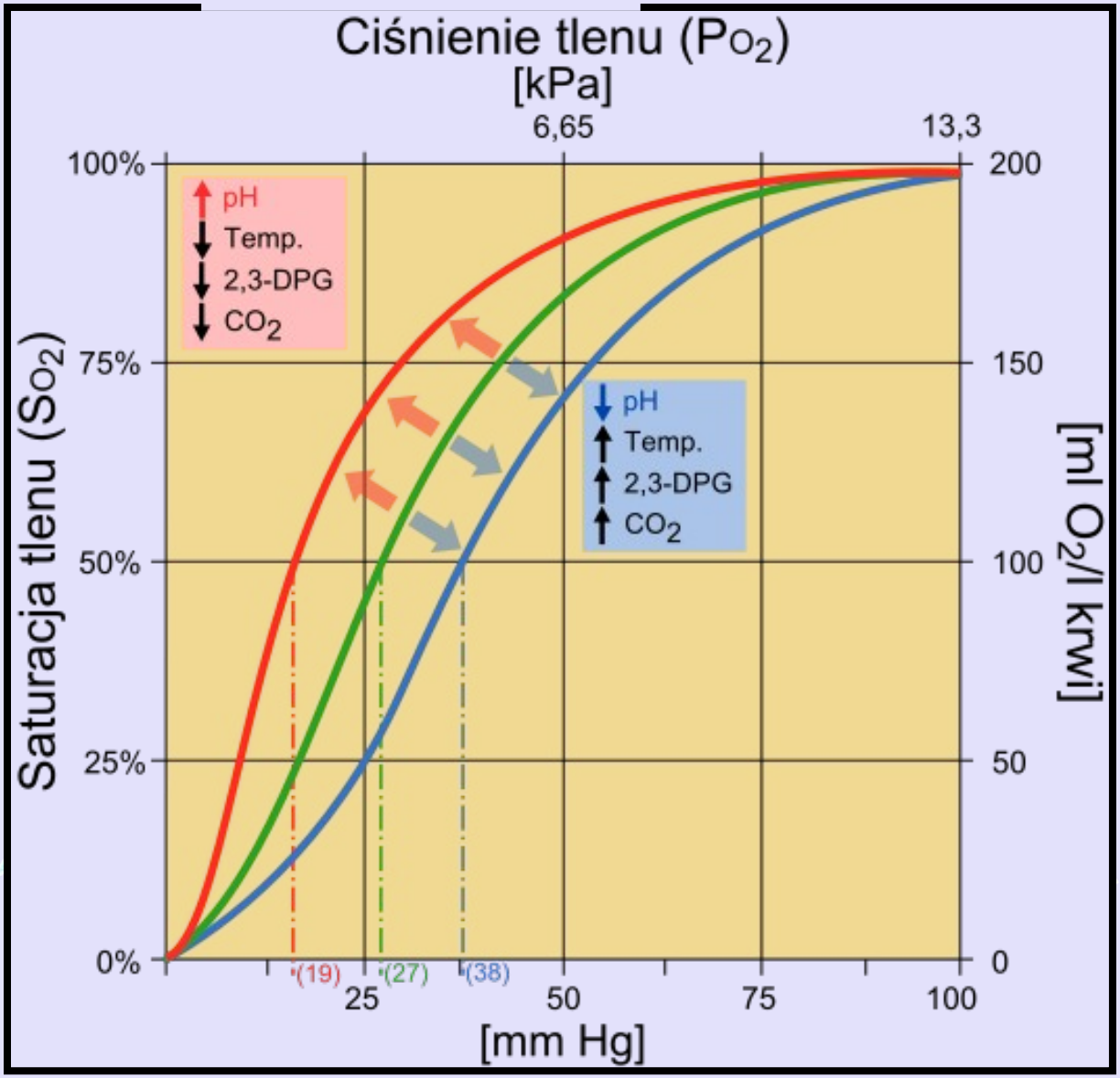


AT LUNGS LEVEL

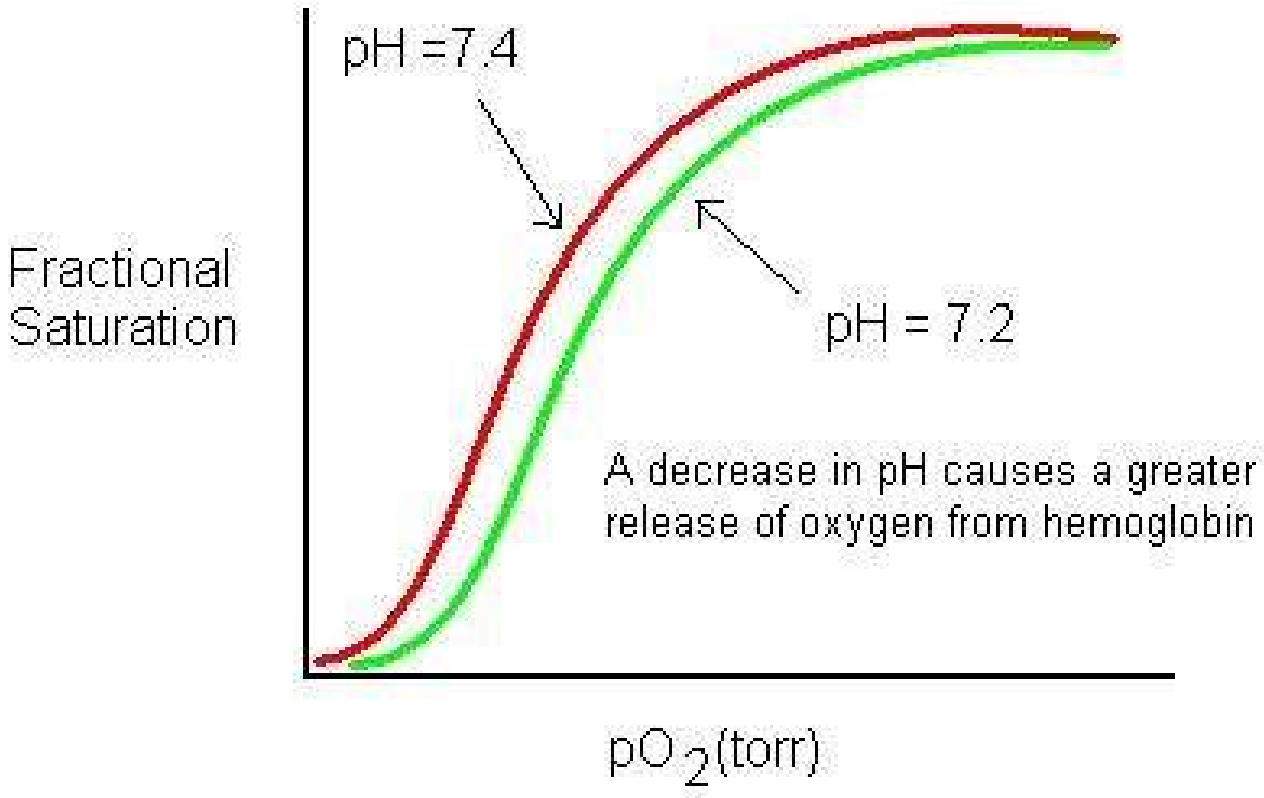


BOHR EFFECT

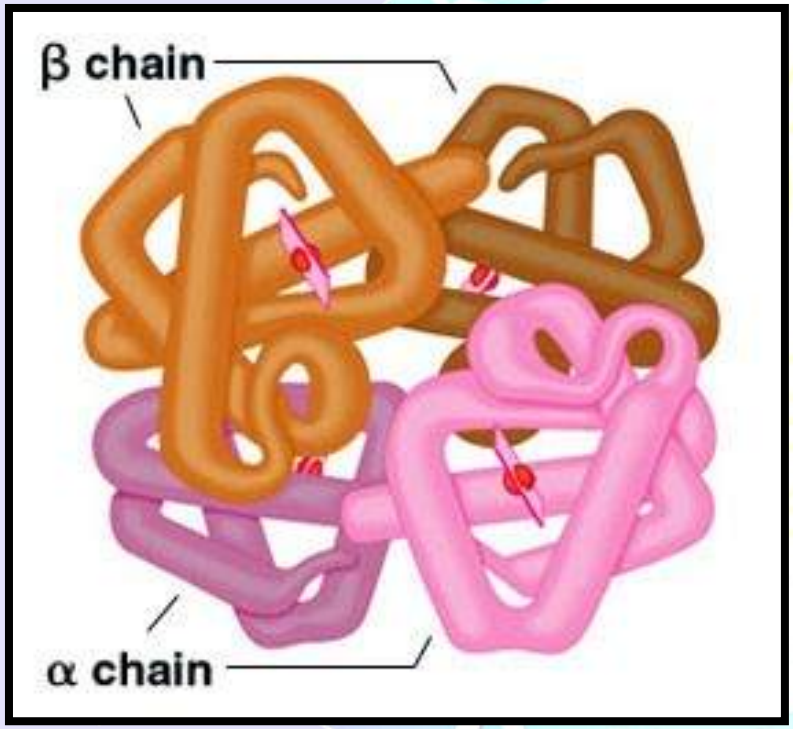
- The release of O_2 from hemoglobin is enhanced when pH is lowered or there is increased pCO_2
- Both result in a decreased O_2 affinity of Hb and a shift to the right in the O_2 dissociation curve
- Raising the pH or lowering the pCO_2 results in greater affinity for oxygen and a shift to the left in oxygen dissociation curve
- This change in oxygen binding is called **Bohr effect**

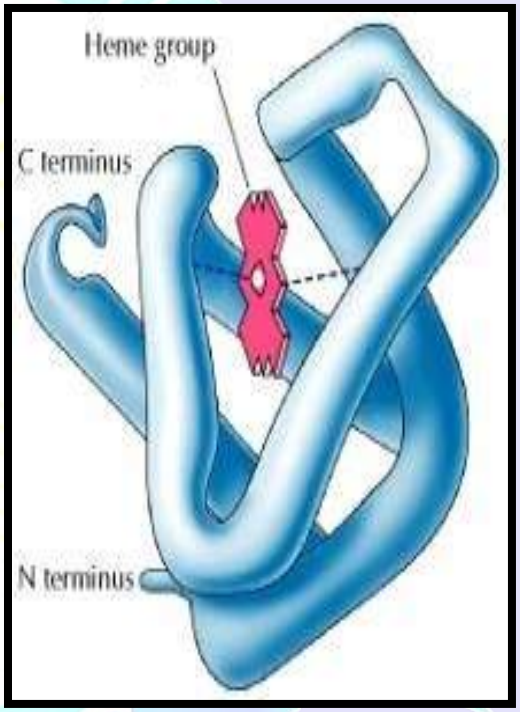

Ciśnienie tlenu (P_{O_2})
[kPa]
6,65
13,3
100%
75%
50%
25%
0%
200
150
100
50
0
pH
Temp.
2,3-DPG
CO_2
pH
Temp.
2,3-DPG
CO_2
Saturacja tlenu (S_{O_2})
[ml O_2/l krwi]
(19)
(27)
(38)
25
50
75
100
[mm Hg]

Effect of pH on hemoglobin



HEMOGLOBIN & MYOGLOBIN

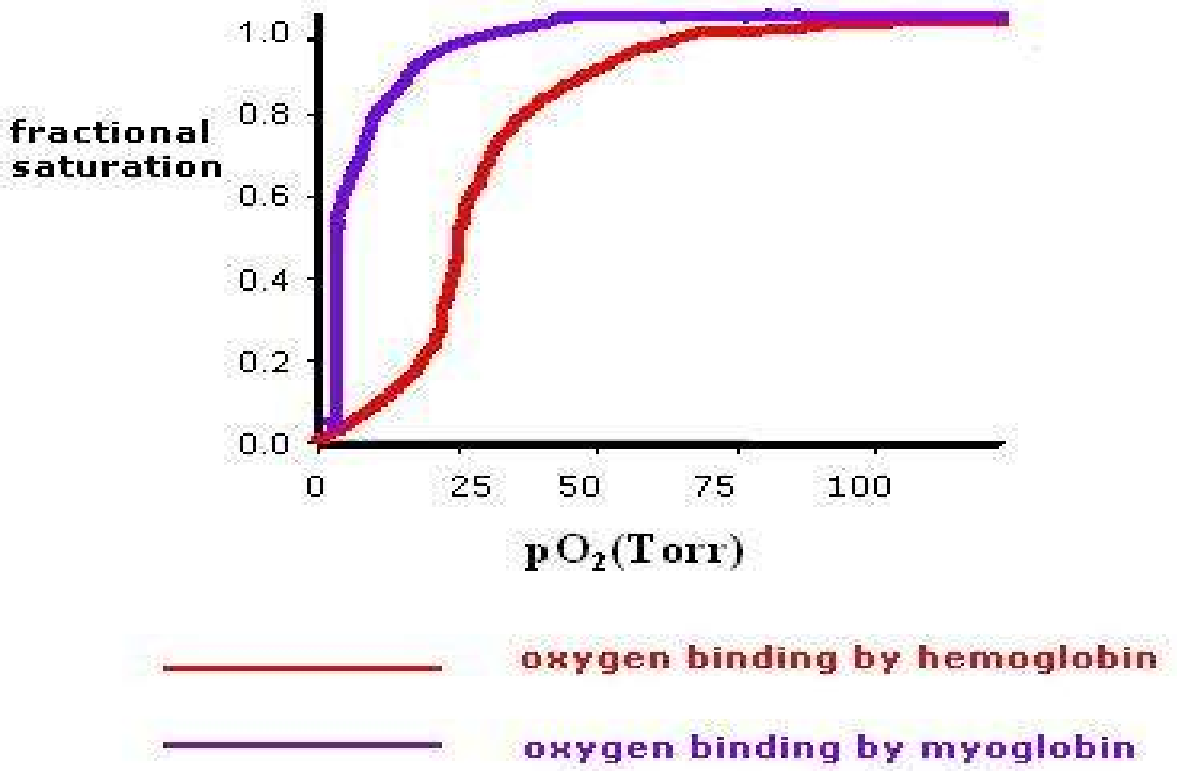




Hemoglobin & Myoglobin

Hemoglobin	Myoglobin
Found in Blood	Found in Heart and skeletal muscles
Composed of 4 Heme and 4 Globin chains	Composed of 1 Heme and 1 Globin chain
Carrier of Oxygen and Carbon dioxide	Reservoir and Carrier of Oxygen
Higher Oxygen affinity	Lesser Oxygen affinity

Oxygen Dissociation Curves



REFERENCES

- GUYTON AND HALL PHYSIOLOGY


Thank You!